

Theory, analysis and applications of entropic lattice Boltzmann model for compressible flows

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Abstract

The entropic lattice Boltzmann method for the simulation of compressible flows is studied in detail and new opportunities for extending operating range are explored. In particular, we address limitations on the maximum Mach number and temperature range allowed for a given lattice. Solutions to both the problems are presented by modifying the original lattices, yet without increasing the number of discrete velocities, and without modification of the numerical algorithm. In order to increase the Mach number, we employ shifted lattices while the magnitude of lattice speeds is increased in order to extend the temperature range. Accuracy and efficiency of the shifted lattices are demonstrated with simulations of the supersonic flow field around a diamond-shaped and NACA0012 airfoil, the subsonic, transonic and supersonic flow field around the Busemann biplane, and the interaction of vortices with a planar shock wave. For the lattices with extended temperature range, the model is validated with the simulation of the Richtmyer-Meshkov instability.

Keywords: Kinetic theory, Lattice Boltzmann method, Compressible flow

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1. Introduction

Predicting behavior of compressible flows is crucial in a great variety of natural phenomena and technological applications, ranging from magnetohydrodynamic and astrophysics all the way down to high speed aerodynamic, turbomachinery and abrasive blasting, to mention a few. While compressible flows are well established by the Navier-Stokes and energy equations, computations remain difficult in practice [1, 2]. Particularly challenging is the interaction of shock waves with turbulence, which requires special treatment by conventional flow solvers. Indeed, higher-order discretization methods used for DNS of turbulent flows can cause Gibbs oscillations, leading to instabilities in the presence of shocks whereas typical methods used to regularize shock computations demonstrate numerical dissipation which degrades accuracy of DNS in regions of high turbulence [2]. Therefore, hybrid schemes combining higher-order methods with (W)ENO schemes through state-of-the-art shock sensors is the preferred approach to the simulation of high-speed compressible flows. The main drawback of these approaches, however, is the complexity of resulting numerical schemes, requiring flow-specific parameter tuning. Thus, development of efficient, high-fidelity, robust DNS solvers which can treat *seamlessly* any type of compressible turbulent sub-, trans- and supersonic flow is a topic of paramount importance.

Recently, the lattice Boltzmann method (LBM) gained attention as an alternative for complex flows simulations including turbulence, micro-scale flows, porous media, multiphase flows, relativistic hydrodynamics, soft glasses and beyond [3]. In LBM, one solves numerically a set of fully discrete (in time-space-velocity) kinetic equations for the populations $f_i(\mathbf{x}, t)$, designed to reproduce, in the hydrodynamic limit, the target equations of fluid dynamics. Populations are associated with discrete velocities $\mathbf{v}_i$ which fit into a regular spatial lattice with nodes $\mathbf{x}$ at time t . LBM is realized by an explicit “stream-along-links-and-equilibrate-at-nodes” algorithm which achieves both computational efficiency and numerical accuracy.

However, realizations of high Reynolds and high Mach number flows with the LBM were limited by numerical instabilities at low viscosity and restricted accuracy of standard lattices. The former confined the range of applications to low Reynolds numbers, while the latter precluded higher flow velocity necessary to achieve subsonic, transonic and supersonic conditions. Among many suggestions [4, 5, 6, 7, 8, 9], the ELBM for incompressible flows [10, 11, 12], disentangled the LBM from the low Reynolds regime by restoring the second law of thermodynamics (Boltzmann’s H theorem). This basic physical requirement rendered the method nonlinearly stable

numerically.

It is possible to increase the flow velocity on standard LB lattices, supplemented by correction terms to improve isotropy. The main drawback is that evaluation of gradients of velocity and temperature at each lattice site becomes necessary [13, 14]. Suitable for thermal and moderate high Mach number flows, such an approach surrenders one of the main advantages of LB methods, namely locality of computations which affects not only computational efficiency and simplicity, but also in part the accuracy of the method.

An alternative formulation of the LBM for thermal flows is to use two sets of populations. The first lattice is used for mass and momentum computations (recovering continuity and momentum equations), while the second lattice is used for the temperature dynamics by conserving the internal energy [15]. However, this coupling in the formulation of [15] is nonlocal, increasing the complexity of the model. In particular, physical effects such as viscous heating do not emerge naturally and need to be introduced by forcing terms. More recently, total energy was chosen instead of the internal energy as the conserved quantity on the second lattice [16, 17, 18] to avoid non-locality of computations. However, this approach still cannot be used for simulation of compressible flows at moderate Mach numbers or acoustics owing to the lack of sufficient isotropy of the first lattice. This drawback is partially eliminated in [19] by introducing correction terms similar to [14]; however the main disadvantages of [14] persist.

In a third approach, correction terms and a second set of populations are avoided by using lattices with more discrete velocities (multi-speed or higher-order lattices), required to adequately represent the moment system necessary to obtain the full thermal Fourier-Navier-Stokes equations. Early attempts to introduce higher-order lattices were based on discretizing the Maxwell-Boltzmann distribution on the roots of Hermite polynomials [20]. While this route of higher-order Gauss-Hermite quadratures promised a systematic derivation of new complete Galilean-invariant LB models, the corresponding discrete velocities cannot be fitted into a regular space-filling lattice. Thus, one of the most important advantages of the LB methods, the exact space discretization of the advection step, is lost with the Gauss-Hermite quadrature-based off-lattice models. Regular space-filling lattices were obtained based on a quadrature representation of the equilibrium moments not necessarily Gauss-Hermite [21, 22, 23, 24, 25, 26], where the nodes of the quadrature are chosen empirically in such a way as to reflect higher-order isotropy. However, none of these derivations could prove to construct an accurate numerical scheme without employing some stabilization technique. Stabilization techniques like equilibration, for example,

succeeded in the simulation of compressible flows as in [27]. An alternative way of deriving space-
 65 filling higher-order lattices was set by the development of the theory of admissible higher-order
 lattices [28, 29], which enabled systematic derivation of lattices of any size, being at the same
 time entropy supporting and guaranteeing the stability intrinsic to the lattice. First application
 of admissible higher-order lattices for turbulent flows simulations was demonstrated in [30], while
 more recently, the same has been applied to thermal flows in [31].

70 2. Compressible lattice Boltzmann model

2.1. Discrete kinetic equations

In the compressible entropic lattice Boltzmann method we solve two sets of kinetic equations for
 the populations f_i and g_i :

$$f_i(\mathbf{x} + \mathbf{v}_i, t + 1) - f_i(\mathbf{x}, t) = \Omega_i^f, \quad (1)$$

$$g_i(\mathbf{x} + \mathbf{v}_i, t + 1) - g_i(\mathbf{x}, t) = \Omega_i^g, \quad (2)$$

with the following conservation laws:

$$\rho = \sum_{i=1}^n f_i, \quad (3)$$

$$\rho u_\alpha = \sum_{i=1}^n f_i v_{i\alpha}, \quad (4)$$

$$2\rho E^{tot} = 2\rho C_v T + \rho u^2 = \sum_{i=1}^n f_i v_i^2 + \sum_{i=1}^n g_i, \quad (5)$$

where ρ is the mass density, u_α is the flow velocity, E^{tot} is the total energy, T is the temperature
 and C_v is the heat capacity at constant volume. When the DdQ7^d lattice described below in
 Sec. 2.2 and the equilibrium described below in Sec. 2.3 are employed, the equations for the
 populations f_i recover, in the hydrodynamic limit, the continuity and momentum equations [32].
 In the case of a monoatomic ideal gas, with adiabatic exponent fixed to $\gamma = \frac{D+2}{D}$, where D is
 the spatial dimension, only the first equation (1) suffices and can be employed to recover also
 the temperature equation, by satisfying the conservation of the total energy,

$$2\rho E^{tr} = \rho D T + \rho u^2 = \sum_{i=1}^n f_i v_i^2. \quad (6)$$

In order to overcome the fixed adiabatic exponent constraint arising from Eq. (1) alone, several approaches have been proposed in the literature [33, 34]; however, they are limited to models with a prescribed closed-form expression of the equilibrium. As it will be explained below, the equilibrium employed in the present model does not offer a closed-form solution, therefore we employ an alternative way, which follows the idea of Rykov’s kinetic model for diatomic molecules [35] and which was already presented for the lattice Boltzmann method in [36]. To that end, a second set of populations g_i Eq. (2) is introduced; these populations carry the energy related to internal degrees of freedom (rotational and vibrational) and thus enables a variable γ .

Left hand side (LHS) of Eqs. (1) and (2) is the standard LB shift operator, representing the advection of populations, while the right hand side (RHS) is the collision operator representing the relaxation of the populations towards equilibrium. No modification of the LB scheme for time and space discretization is used in this paper. The collision operator Ω_i^f is modeled with a modified LBGK model, the quasi-equilibrium model [37] combined with the entropic collision [12, 28]:

$$\Omega_i^f = \alpha\beta_1 (f_i^{\text{eq}} - f_i) + 2(\beta_1 - \beta_2) [f_i^* - f_i^{\text{eq}}]. \quad (7)$$

The first term of the RHS of Eq. (7) becomes the conventional LBGK model in case $\alpha = 2$, while it represents the entropic collision when α is computed by satisfying the discrete time H -theorem, i.e. when it is computed as the positive root of the entropy condition:

$$H(f + \alpha(f^{\text{eq}} - f)) = H(f) \quad (8)$$

where H is the entropy function [10],

$$H(f) = \sum_{i=1}^n f_i \ln \left(\frac{f_i}{W_i} \right). \quad (9)$$

For more details about the entropic estimate in the context of the compressible model the reader is referred to [32]. The second term represents relaxation to the quasi-equilibrium state and is employed to modify the Prandtl number. The collision operator for the second set of populations g_i is designed to be consistent with the f_i equations and reads:

$$\Omega_i^g = \alpha\beta_1 (g_i^{\text{eq}} - g_i) + 2(\beta_1 - \beta_2) [g_i^* - g_i^{\text{eq}}]. \quad (10)$$

In both collision operators superscript eq denotes equilibrium states, while superscript $*$ denotes the quasi-equilibrium state. Since equilibrium is a central point in the construction of the compressible model we will detail it in a separate section, Sec. 2.3. The quasi-equilibrium states

need to be chosen depending on the Prandtl number [31], and in particular, for the case $\text{Pr} \leq 1$, the quasi-equilibrium state conserves the centered heat-flux tensor $\bar{\mathbf{Q}}$. It can be written in the compact form as:

$$f_i^* = f_i^{\text{eq}} + W_i \frac{(\bar{\mathbf{Q}} - \mathbf{Q}^{\text{eq}}) : [\mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i - 3T\mathbf{v}_i\mathbf{I}]}{6T^3}, \quad (11)$$

where

$$\bar{\mathbf{Q}} = \sum_{i=1}^n f_i (\mathbf{v}_i - \mathbf{u}) \otimes (\mathbf{v}_i - \mathbf{u}) \otimes (\mathbf{v}_i - \mathbf{u}). \quad (12)$$

Quasi-equilibrium g_i^* populations are written consistently with the f_i^* populations and read:

$$g_i^* = g_i^{\text{eq}} + \frac{W_i (\bar{\mathbf{q}} - \mathbf{q}^{\text{eq}}) \cdot \mathbf{v}_i}{T}, \quad (13)$$

where $\bar{\mathbf{q}}$ is the contracted centered heat-flux tensor associated to the internal degrees of freedom of the gas:

$$\bar{\mathbf{q}} = \sum_{i=1}^n g_i (\mathbf{v}_i - \mathbf{u}). \quad (14)$$

80 In the above expressions $W_i = W_i(T)$ are the lattice-dependent weights which will be detailed later. While the present model is generic in nature, the details regarding the construction of the equilibria and the choice of the lattice still need to be exposed.

2.2. Lattices

In order to simulate thermal or fully compressible flows, we employ here higher order lattices.

85 We consider only lattices belonging to the theory of admissible higher-order lattices, presented for the first time in [28, 29, 38], and for which a summary is provided in Table 1 for lattices until $n = 11$ discrete velocities (in one-dimension). The table 1 shows, for each discrete velocity

n	V_n	$T_{\min}$	$T_{\max}$	T_0	$\theta_{\min}$	$\theta_{\max}$	$\frac{T_{\max}}{T_{\min}}$
3	$\{0, \pm 1\}$	0.0	1.0	1/3	-1	2	∞
5	$\{0, \pm 1, \pm 3\}$	1/3	3.0	$1 \pm \sqrt{2/5}$	-0.09308	0.83772	9
7	$\{0, \pm 1, \pm 2, \pm 3\}$	$1 - \sqrt{2/5}$	$1 + \sqrt{2/5}$	0.69795	-0.47340	1.33892	4.44
9	$\{0, \pm 1, \pm 2, \pm 3, \pm 5\}$	0.69795	2.88131	0.75608, 2.17538	-0.07688	0.32451	4.13
11	$\{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$	0.75608	2.17538	1.06279	-0.28859	1.04685	2.88

Table 1: One-dimensional lattices with odd number of velocities n . In order: lattice velocities set V_n , minimal temperature $T_{\min}$, maximal temperature $T_{\max}$, reference temperature T_0 , minimal reduced temperature $\theta_{\min}$, maximal reduced temperature $\theta_{\max}$ and temperature ratio $T_{\max}/T_{\min}$.

set $V_n = \{0, \pm 1, \dots\}$, the corresponding lattice links size, the temperature range, the reference temperature T_0 , the reduced temperature $\theta = T/T_0 - 1$ range and the ratio between maximum and minimum temperature T_{max}/T_{min} at $\mathbf{u} = 0$. Outside the temperature range, the weights and hence the equilibrium populations become negative and the scheme may become unstable. For this reason, the ratio T_{max}/T_{min} can be an important criterium to determine which lattice can be employed depending on the problem at hand. Another criterium is the error in the third- and fourth-order equilibrium moments from their Maxwell-Boltzmann (continuous) counterparts. As the number of velocities is increased, for a given Mach number, the errors are reduced, as will be shown in the next section, Sec. 2.3. It has been shown in previous works [39, 32], that in order to simulate compressible flows the minimum number of velocities is $n = 7$. With this lattice, the errors in the third- and fourth-order moments are kept small until a moderately supersonic Mach number, and the temperature range is sufficient to guarantee positivity of the populations for moderate temperature jumps across shock waves. For these reasons, from now on, the reference lattice for compressible flows will be the DdQ7^d, i.e. the D -dimensional lattice constructed from the velocity-set V_7 , where D is the space dimension. For this lattice, the one-dimensional weights reads:

$$\begin{aligned}
w_0 &= \frac{(36 - 49T + 42T^2 - 15T^3)}{36}, \\
w_{\pm 1} &= \frac{T(12 - 13T + 5T^2)}{16}, \\
w_{\pm 2} &= \frac{T(-3 + 10T - 5T^2)}{40}, \\
w_{\pm 3} &= \frac{T(4 - 15T + 15T^2)}{720},
\end{aligned} \tag{15}$$

while in higher dimensions, the weight W_i of each discrete velocity $\mathbf{v}_i$ in the natural Cartesian reference frame, $v_i = (v_{ix}, v_{iy}, v_{iz})$, is the algebraic product of the corresponding one-dimensional weights,

$$W_i = w_{ix} w_{iy} w_{iz}. \tag{16}$$

In Figure 1, the DdQ7^d lattice is shown for the two-dimensional case, i.e. with $n = 49$ velocities, D2Q49.

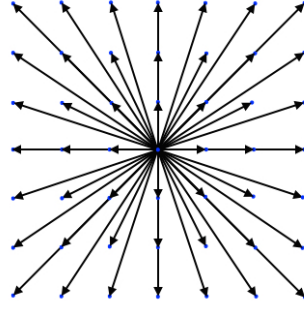


Figure 1: Visualization of the D2Q49 lattice.

2.3. Equilibrium

2.3.1. Equilibrium construction

Here we proceed with the definition of equilibrium which is derived by minimization of the entropy function H Eq. (9). Minimization of H is carried out under constraints of local conservation of mass ρ , momentum $\rho\mathbf{u}$ and translational energy E_{tr} :

$$\{\rho, \rho\mathbf{u}, E^{tr}\} = \{\rho, \rho\mathbf{u}, D\rho T + \rho\mathbf{u}^2\} = \sum_{i=1}^n \{1, \mathbf{v}_i, v_i^2\} f_i(\rho, \mathbf{u}, T). \quad (17)$$

The minimization problem is solved with the method of the Lagrange multipliers (LM), leading to:

$$\delta H + \delta(\chi\rho + \boldsymbol{\zeta} \cdot (\rho\mathbf{u}) + \lambda\rho E_{tr}) = 0, \quad (18)$$

where $\chi(\mathbf{u}, T)$, $\boldsymbol{\zeta}(\mathbf{u}, T)$ and $\lambda(\mathbf{u}, T)$ are Lagrange multipliers corresponding to conservation of mass, momentum and translational energy, respectively. The formal solution to the minimization problem reads:

$$f_i^{eq} = \rho W_i \exp(\chi + \boldsymbol{\zeta} \cdot \mathbf{v}_i + \lambda v_i^2). \quad (19)$$

The closed form of f_i^{eq} is computed by applying the conservation laws (17) to the equilibrium populations (19). This results in $D+2$ equations for $D+2$ unknown LM. For the standard lattice ($v_i = \{0, \pm 1\}$), this problem was solved analytically in [12]. However for higher-order lattices, say the D1Q5 and D1Q7 lattice (and the respective in higher dimensions) and with energy conservation, this leads to a system of $D+2$ algebraic equations of the order $O = 9D+3$ (for maximum lattice velocity $v_3 = 3$); thus ruling out any possibility of computing exact solution. We suggest here direct numerical evaluation of LM using the rapidly converging Newton-Raphson method. Our simulations shows that such an approach converges to an accurate solution (with

error of order $O(10^{-14})$) within five iterations when LM from previous time step are employed as initial guess for the Newton-Raphson solver. Note that the equilibrium g_i^{eq} need not to be computed by another Newton-Raphson iteration since, after evaluation of f_i^{eq} , we set

$$g_i^{\text{eq}} = (C_v - D/2)Tf_i^{\text{eq}}. \quad (20)$$

At a first glance, the proposed evaluation of the equilibrium at every node and every time-step appears computationally demanding. In the following, we analyze how the numerical computation
 110 of the equilibrium impacts the overall computational load of the algorithm.

2.3.2. Computational costs of numerical equilibrium

In order to estimate the performance of the accurate numerical equilibrium, and to compare to a standard polynomial equilibrium evaluation, we report here the computational time required to run a constant homogenous advection flow at Mach number $\text{Ma}_a = 0.05$. It must be noted that these computational times are evaluated for the full implementation (advection and collision) of the LBGK algorithm with periodic boundary conditions. Considered are the accurate numerical equilibrium Eq. (19), and a fourth order approximation of the equilibrium in polynomial form. The fourth order approximation is theoretically applicable only if n is “large enough” i.e. if the lattice velocity space can accommodate all higher order moments required for recovering the compressible flow hydrodynamic equations. The fourth order approximation is given by:

$$f_i^{\text{eq}} = \rho W_i \left(1 + \frac{u_\alpha v_{i\alpha}}{T} + \frac{u_\alpha u_\beta X_{i\alpha\beta}}{2T^2} + \frac{u_\alpha u_\beta u_\gamma Y_{i\alpha\beta\gamma}}{6T^3} + \frac{u_\alpha u_\beta u_\gamma u_\mu Z_{i\alpha\beta\gamma\mu}}{24T^4} \right), \quad (21)$$

with

$$X_{i\alpha\beta} = v_{i\alpha}v_{i\beta} - T\delta_{\alpha\beta}, \quad (22)$$

$$Y_{i\alpha\beta\gamma} = X_{i\alpha\beta}v_{i\gamma} - v_{i\alpha}T\delta_{\beta\gamma} - v_{i\beta}T\delta_{\alpha\gamma}, \quad (23)$$

$$Z_{i\alpha\beta\gamma\mu} = Y_{i\alpha\beta\gamma}v_{i\mu} - X_{i\alpha\beta}\delta_{\gamma\mu}T - X_{i\alpha\gamma}\delta_{\beta\mu}T - X_{i\beta\gamma}\delta_{\alpha\mu}T. \quad (24)$$

One should notice, that this approximation can be employed for the simulation of energy conserving flows but only for moderately low Mach number flows, due to truncation errors and non-positivity of the populations, as it will be discussed in Sec. 2.3.4.

115 In Table 2, we report the computational time measurements normalized by the fastest evaluation, namely the polynomial evaluation on the DdQ3^d lattice, for the two different equilibrium

evaluations, for the three different lattices $DdQ3^d$, $DdQ5^d$ and $DdQ7^d$, and in two- and three-dimensions.

	$DdQ3^d$	$DdQ5^d$	$DdQ7^d$
<i>2D</i>			
Polynomial	1	4.12	8.13
Numerical	2.02	4.83	8.74
<i>3D</i>			
Polynomial	1	8.91	25.66
Numerical	1.58	10.83	29.77

Table 2: Computational time for the LBGK algorithm with evaluation of the equilibrium using the polynomial and the numerical method, for three lattices $DdQ3^d$, $DdQ5^d$ and $DdQ7^d$, in two- and three-dimensions. Computational time is normalized by the fastest implementation, i.e. with polynomial form and on the $DdQ3^d$ lattice.

Firstly, it is interesting to notice that the computational time required by the numerical equilibrium on every lattice is not more than the double the computational time of the polynomial equilibrium. Once the number of speeds is increased, the difference between the numerical equilibrium and the polynomial form decreases, in both two- and three-dimensions. This is explained by the fact that numerical equilibrium computes $(D + 2)$ Lagrange multipliers, independently of the number of speeds n , while for the polynomial form, n fourth-order polynomials need to be computed. Table 2 clearly shows that the computational cost depends mainly on the number of discrete lattice velocities, while the type of equilibrium plays a secondary role. This analysis shows clearly that the employment of the accurate form of the equilibrium is not heavily penalizing the performance of the simulation with respect to a polynomial form of the equilibrium.

A general comment on the computational cost of the present model is in order here. As just seen, the computational cost of the present compressible model is one to two orders of magnitude higher (in 2 and 3 dimensions, respectively) than its incompressible counterpart, the $DdQ3^d$. However, for compressible (or high Mach number) simulations the time step $\delta t U/L$, where U and L are characteristic velocity and length, is in general one to two orders of magnitude higher than for the incompressible simulations, since the time step is directly proportional to the characteristic Mach number of the simulations. This implies that no significant increase of computational cost

may be observed in present simulations with respect to incompressible LBM simulation. Further it must be pointed out that incompressible ELBM simulations were shown to be one to two order of magnitude faster than DNS simulations using state-of-the-art techniques, for both periodic
140 flows [30] and flow in complex geometries [40].

It is also interesting to analyze the influence of computational load with the variation of the flow field (ρ , $\mathbf{u}$ and T) for the numerical evaluation of the equilibrium. In fact, since the Newton-Raphson solver for the equilibrium employs previous time step LM as initial guess, one would
145 expect that a high time variability of the flow could influence the performance. We propose here to measure the influence of the time variability of the flow field with the simulation of compressible decaying isotropic turbulence similar to the one presented in [39], for different initial turbulent Mach numbers Ma_t . The initial turbulent Mach number is considered here as the quantitative measure of the time variability of the flow. In Table 3 we show the influence of
150 the time-variability of the flow (in terms of Ma_t) by measuring the computational time of the various decaying turbulence simulations. The lattice employed for the simulations is the D3Q7³.

Ma_t	0	0.1	0.2	0.3	0.4	0.5	0.6
t/t_0	1.000	1.047	1.023	1.059	1.062	1.070	1.067

Table 3: Computational time dependence on turbulent Mach number Ma_t . t_0 is the time measured for a very low turbulent Mach number near to $\text{Ma}_t \simeq 0$.

As one can notice, the computational time for the simulation increases only slightly with increasing turbulent Mach number. In fact, the computational time is increased at most by 7%,
155 when the turbulent Mach number is quite high (the local Mach number may reach values above $\text{Ma}_{\text{loc}} \simeq 2.0$). This test also includes the uneven load balancing experienced in the simulation due to varying local Mach numbers, i.e. the fact that in some processors the flow field varies much less or much more than in other processors.

2.3.3. Numerical equilibrium accuracy

In this section we analyze the numerical equilibrium accuracy in terms of deviation of the pertinent moments from Maxwell-Boltzmann counterparts:

$$\epsilon_m = \frac{M_m^{eq} - M_m^{MB}}{M_m^{MB}}. \quad (25)$$

Superscript MB denotes the moments computed with the continuous Maxwell-Boltzmann distribution of a general moment of order m . In Fig. 2, we compare the D2Q5² and D2Q7² lattices for the errors in the equilibrium heat flux q_x^{eq} and equilibrium contracted fourth order moment R_{xx}^{eq} at various Mach numbers. Two types of Mach number are here defined: Ma_t stands for turbulent Mach number and Ma_a is the advection Mach number. The equilibrium is computed for a Mach number Ma_a in the x -direction plus a Mach number Ma_t superimposed in both the x - and y -directions. In the same plot, for completeness, the errors of the eleven D2Q11² lattice are also shown.

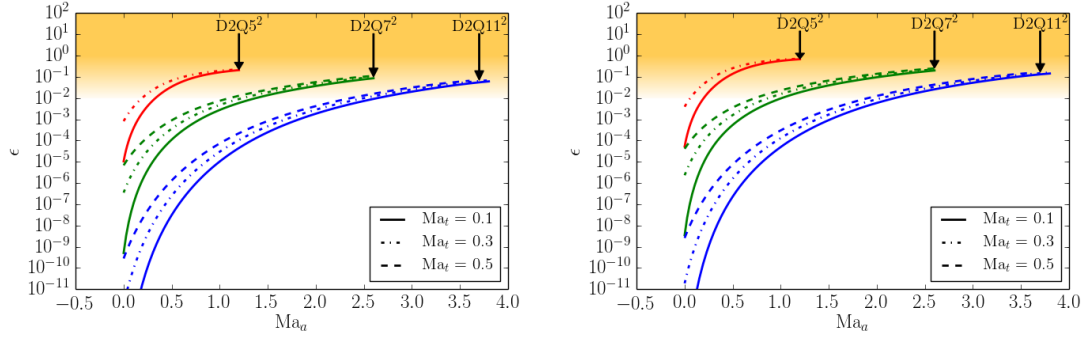


Figure 2: Deviation of the contracted, x-component of third order moment $q_x^{eq} = \sum_{i=1}^n v_{ix} v_i^2 f_i^{eq}$ (left) and of the xx-component of fourth-order moment $R_{xx}^{eq} = \sum_{i=1}^n v_{ix} v_{ix} v_i^2 f_i^{eq}$ (right), from the Maxwell-Boltzmann values, as a function of turbulent and advection Mach number Ma_t and Ma_a respectively, for three different lattices: D2Q5², D2Q7² and D2Q11².

Important to notice is that the aforementioned moments need to be matched as accurately as possible by the given lattice, in order to recover the correct thermo-hydrodynamic limit. One can notice that the errors in the moments are small until advection Mach number of about $Ma_a \leq 1.2$ for the D2Q7² lattice, while for the D2Q5² lattice the errors grow rapidly already in the subsonic regime. This implies that in order to perform supersonic simulations the minimum number of lattice velocities is $n = 7^D$ in D dimensions. For this reason, from now on, we will

consider the D2Q7² lattice as the minimal (among the lattices built by a direct product of the one-dimensional velocity sets) in order to perform compressible flow simulations.

In Fig. 3, the same errors as for Fig. 2 are plotted as a function of the reduced temperature θ , for two different lattices D2Q5² and D2Q7² at different Mach numbers. Errors are shown within the reduced temperature range appropriate for the lattice as discussed in Section 2.2. First of all, one can notice, similarly as before, that by increasing the number of velocities, the

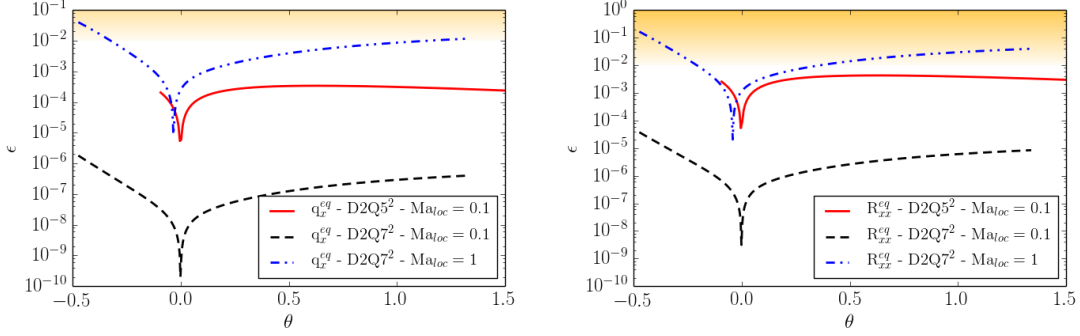


Figure 3: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of the reduced temperature θ for two different lattice sizes: D2Q5² at local Mach $Ma_{loc} = 0.1$ and D2Q7² at local Mach $Ma_{loc} = 0.1$ and $Ma_{loc} = 1$.

errors decrease by order of magnitude, and the same is valid even when moving away from the reference temperature. Moreover, we can see that by increasing the Mach number, the errors when deviating from reference temperature are amplified. However, considering that typical medium strength shock experience a reduced temperature jump of $\Delta\theta \simeq 0.3$, we can state that even when increasing the Mach number the deviations from the equilibrium are small enough to perform supersonic flow simulations with shocks on the DdQ7^d lattice.

2.3.4. Equilibria positivity

In order to conclude this section we point at a remarkable property of the numerical equilibrium employed in this work. For the lattice DdQ7^d discussed in Sec. 2.2, we provide a quantitative impression of the difference between evaluating the equilibrium with accurate numerical procedure and with a polynomial approximation. In Figure 4 we plot, for the one-dimensional case D1Q7, populations f_0^{eq} , f_2^{eq} and f_{-2}^{eq} evaluated in three different ways as a function of the Mach number Ma : a third and a fourth order approximation (in terms of $\mathbf{u}$) of equilibrium (19), and the accurate equilibrium (Newton-Raphson evaluation). It is clear that as the Mach

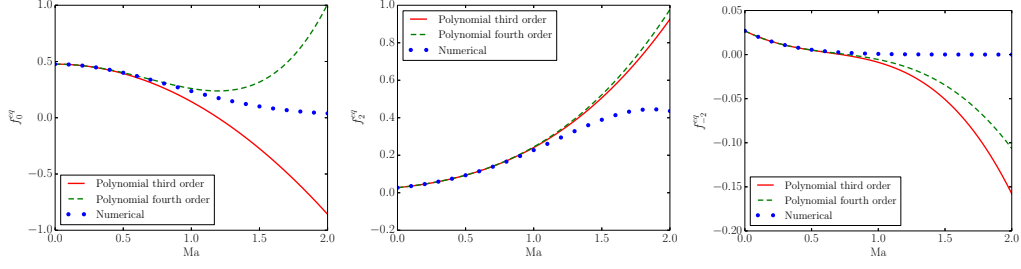


Figure 4: Equilibrium population f_0^{eq} , f_2^{eq} and f_{-2}^{eq} evaluated as a function of Mach number with three different methods: third order and fourth order polynomial approximation in velocity and exact numerical evaluation.

number is increased, especially after the Mach number reaches the sonic value, the polynomial approximations start to diverge from the exact solution to the minimization problem, leading to accumulation of errors. Moreover, an increase of the degree of approximation, from a third order in velocity to a fourth order in velocity, has only a minor influence on the convergence to the correct solution. Important to notice is also, that after a certain value of the Mach number the populations evaluated with a polynomial approximation become negative, violating the positivity constraints imposed by the construction of the equilibrium. From Figure 4, it is possible to conclude that numerical evaluation of the equilibrium is not just more accurate, but it is also unavoidable for compressible lattice Boltzmann simulations.

However, the constraint of positivity on the entropic equilibria comes along with a cost: as seen in Section 2.3.3, the equilibrium moments starts to deviate from their continuous Maxwell-Boltzmann counterparts when the Mach number is increased above a certain limit or/and the temperature deviates from reference temperature T_0 . In general, from the above analyses of the errors as a function of Mach number Ma and reduced temperature θ , we observe that the $DdQ7^d$ lattice has all the minimal characteristics required to run compressible flows simulations, i.e. accuracy at relatively high Mach numbers and high temperature range guaranteeing positivity of the populations. In Section 3 we extend the operation window in terms of Mach number and temperature range by simply modifying the lattice configuration and without increasing the number of populations.

2.4. Thermo-Hydrodynamic Equations

In [32], the thermo-hydrodynamic limit of Eq. (1) together with (2) is derived, recovering the Fourier-Navier-Stokes equations as follows:

$$\partial_t \rho + \partial_\alpha (\rho u_\alpha) = 0, \quad (26)$$

$$\partial_t u_\alpha + u_\beta \partial_\beta u_\alpha = -\frac{1}{\rho} \partial_\alpha (\rho T) - \frac{1}{\rho} \partial_\beta \Pi_{\alpha\beta}, \quad (27)$$

$$\partial_t T + u_\alpha \partial_\alpha T = -\frac{1}{\rho C_v} \rho T (\partial_\alpha u_\alpha) - \frac{1}{\rho C_v} (\partial_\alpha u_\beta) \Pi_{\alpha\beta} + \frac{1}{\rho C_v} \partial_\alpha (\kappa \partial_\alpha T), \quad (28)$$

where, the stress tensor is defined as,

$$\Pi_{\alpha\beta} = -\mu \left(S_{\alpha\beta} - \frac{2}{D} \partial_\gamma u_\gamma \delta_{\alpha\beta} \right) + \xi \partial_\gamma u_\gamma \delta_{\alpha\beta}, \quad (29)$$

with the strain rate tensor,

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha. \quad (30)$$

Expressions for dynamic viscosity μ , bulk viscosity ξ , thermal conductivity κ and adiabatic exponent γ_{ad} are given by,

$$\mu = \begin{cases} \left(\frac{1}{2\beta_1} - \frac{1}{2} \right) \rho T, & \text{if } \text{Pr} \leq 1, \\ \left(\frac{1}{2\beta_2} - \frac{1}{2} \right) \rho T, & \text{if } \text{Pr} > 1, \end{cases} \quad (31)$$

$$\kappa = \begin{cases} C_p \left(\frac{1}{2\beta_2} - \frac{1}{2} \right) \rho T, & \text{if } \text{Pr} \leq 1, \\ C_p \left(\frac{1}{2\beta_1} - \frac{1}{2} \right) \rho T, & \text{if } \text{Pr} > 1, \end{cases} \quad (32)$$

$$\xi = \left(\frac{1}{C_v} - \frac{2}{D} \right) \mu, \quad (33)$$

$$\gamma_{ad} = \frac{C_p}{C_v}, \quad (34)$$

and the expression for the Prandtl number becomes:

$$\text{Pr} = \begin{cases} \frac{(1-\beta_1)\beta_2}{(1-\beta_2)\beta_1}, & \text{if } \text{Pr} \leq 1, \\ \frac{(1-\beta_2)\beta_1}{(1-\beta_1)\beta_2}, & \text{if } \text{Pr} > 1. \end{cases} \quad (35)$$

3. Extension of operation domain

As outlined in the previous section, the accuracy for a given lattice is restricted in terms of deviation of a local Mach number and of the temperature from their reference values (reference Mach 0, and reference temperature T_0). In this section, we address this issue by retaining the same number of discrete velocities. A brut force way to overcome these issues would be, in fact, to increase the number of lattice discrete velocities. For example, when going from the $DdQ7^d$ lattice to the $DdQ11^d$ lattice, the deviation from Maxwell-Boltzmann moments reduces; between an advection Mach number of $Ma_a = 1$ and $Ma_a = 2$ the reduction of errors is one to two orders of magnitude (see Figure 2). However, as also demonstrated in Table 2, the computational cost increases at least linearly with the number of populations, so that the simulation with the $D2Q11^2$ and especially the $D3Q11^3$ would quickly become prohibitively demanding. In the next two sections, we suggest how the Mach number can be increased and how the temperature range can be extended without enlarging the number of velocities. To that end, the lattice $DdQ7^d$ is considered, as it has already been benchmarked for compressible flows simulations [39, 32].

3.1. Shifted lattices

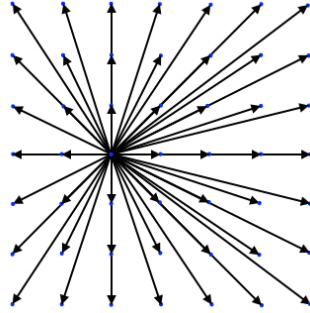


Figure 5: Visualization of the D2Q49 lattice (left) and the corresponding shifted lattice with $U = 1$ (right).

We follow [41] in order to extend the Mach number range. This method is particularly useful when the mean flow has a preferential direction, for example, in the simulation of supersonic flow around obstacles, or simulations with stationary shocks and mean cross-flow like the simulation of shock-vortex interaction or shock-turbulence interaction.

We wish to challenge the notion of a reference frame at rest; instead, we prefer to define the equilibrium in a frame moving, say, with the velocity U in the x -direction. This is achieved by

considering a *shifted* discrete velocity set V'_n with the velocities

$$v'_i = v_i + U, \quad v_i \in V_n, \quad (36)$$

and finding the shifted weights $w_i(U, T)$ by matching the moments of the Maxwellian computed at velocity U and temperature T , $f_v^M(U, T)$:

$$f_v^M(U, T) = \sqrt{\frac{1}{2\pi T}} \exp \left\{ -\frac{(v - U)^2}{2T} \right\}. \quad (37)$$

For symmetric lattices the weights are found by solving the following system of equations ($m = 0, \dots, n - 1$)

$$\langle w_i(0, T) v_i^m \rangle = \langle f_v^M(0, T) v^m \rangle, \quad (38)$$

where operator $\langle \dots \rangle$ stands for summation over discrete velocity index i for the discrete case (left) or integration over the continuous velocity v (right). Instead, for shifted weights $w_i(U, T)$ we have:

$$\langle w_i(U, T) [v_i + U]^m \rangle = \langle f_v^M(U, T) v^m \rangle. \quad (39)$$

Key observation of [41] is that the weights $w_i(U, T)$ are Galilean invariant: For any set V_n and for any shift U we have

$$w_i(U, T) = w_i(0, T). \quad (40)$$

The proof of (40) follows immediately from Galilean invariance of the moments of the Maxwellian: Using binomial theorem and Eq. (38), we write for $m = 0, \dots, n - 1$,

$$\begin{aligned} \langle f_v^M(U, T) v^m \rangle &= \sum_{k=0}^m \binom{m}{k} U^k \langle f_v^M(0, T) v^{m-k} \rangle \\ &= \sum_{k=0}^m \binom{m}{k} U^k \langle w_i(0, T) v_i^{m-k} \rangle \\ &= \langle w_i(0, T) [v_i + U]^m \rangle. \end{aligned}$$

Hence, Eq. (39) is equivalent to

$$\langle [w_i(U, T) - w_i(0, T)] [v_i + U]^m \rangle = 0. \quad (41)$$

Since the shift U is arbitrary, the result (40) follows.

Galilean invariance of the weights (40) has a number of immediate important implications. First, the construction of the discrete velocities through tensor products in higher dimensions remains

as before. For example, in two dimensions, the shift U in the x -direction corresponds to tensor product $V'_{nx} \otimes V_{ny}$. We shall use this example below. Second, the symmetric integer velocity sets generate space-filling lattices. This is crucial for the realization of the time-marching LB scheme (1). Now, if the shift U is itself integer, the corresponding shifted lattice is also space-filling. For example, with $U = 1$, the shifted set V'_7 becomes $V'_7 = \{-2, -1, 0, 1, 2, 3, 4\}$. The shifted lattice, with the discrete velocities $V'_{7x} \otimes V_{7y}$ is shown in Fig. 5. Next, the weights corresponding to the velocities of the shifted lattice are the same as for the symmetric case: $W_i(U, T) = w_{ix}(0, T)w_{iy}(0, T)$. Moreover, since the weights do not change under a transformation to a co-moving reference frame, the entropy function is Galilean invariant, and is given by Eq. (9). Consequently, the equilibrium populations are form-invariant. They are found by the same minimization techniques as in the familiar symmetric case.

The errors associated with the shifted lattices are shown in Fig. 6, for the same quantities as presented in Fig. 2. Presented are the errors for the lattice with no shift, and with a shift $U = 1$ and $U = 2$. The errors, in Fig. 2, are plotted now as a function of the u_x velocity, in order to show the effect of the lattice shift. When the velocity matches the shift velocity, the errors in the moments vanishes.

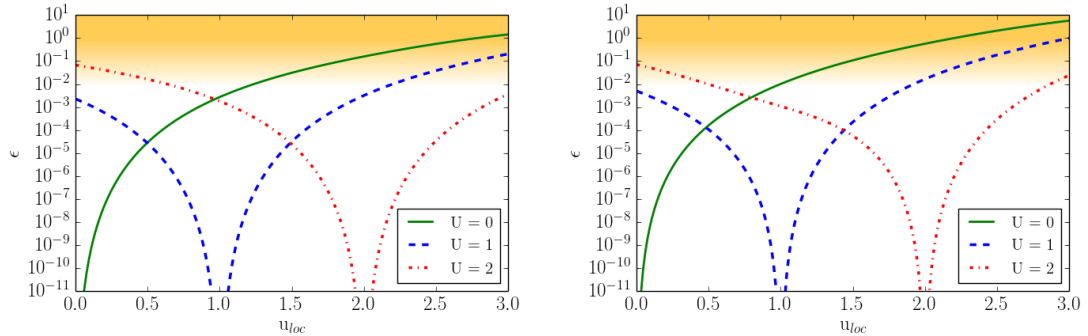


Figure 6: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of local u_x velocity, for non-shifted lattice and for lattices with shift $U = 1$ and $U = 2$.

In order to understand the effect of the shifted lattice, we run test simulations of a vortex advected at different Mach numbers. After a certain Mach number, the non-shifted lattice is expected to feature some numerical errors, due to the lack of Galilean invariance in the moments, as noted in Fig. 6. Due to these numerical errors the vortex is no more advected correctly and starts to

260 deform. The same is expected also for the shifted lattice, but around a higher Mach number (a unity shift in errors). This is demonstrated in Figure 7, for a characteristic vortex Mach number of $\text{Ma}_v = 0.5$ and different advection Mach numbers, $\text{Ma}_a = \{0.3, 0.8, 1.3, 1.8, 2.3\}$, and by means of pressure contours.

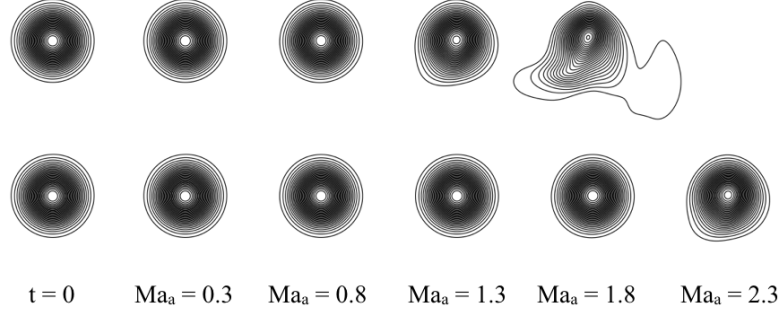


Figure 7: Vortex advection comparison at different advection Mach number. Top row: symmetric lattice. Bottom row: shifted lattice with $U = 1$.

Second order convergence was checked for lattices without and with shifts $U = 1$ and $U = 2$, by simulating the Green-Taylor vortex with a mean advection speed of $U = 0$, $U = 1$, $U = 2$, respectively. The result of the grid convergence study is presented in Fig. 8 and shows no difference between the results of a non- and shifted lattices. We plot the L_2 norm of the error defined as:

$$L_2 = \sqrt{\frac{\sum_i (u_{i,LBM} - u_{i,ref})^2}{\sum_i u_{i,ref}^2}}, \quad (42)$$

265 where $u_{i,LBM}$ is the velocity at sample i with the present simulations, and $u_{i,ref}$ is the velocity of computed from analytical solution.

3.2. Lattices with increased temperature range

270 The range of temperatures supported by the lattice is increased based on the following observation: By considering the velocity set $V_7 = \{0, \pm 1, \pm 2, \pm r\}$, where if $r = 3$ the standard lattice employed for compressible flow in [39] is recovered, we notice that by increasing the longest lattice link to $r = 4$, the temperature range increases significantly, while the lattice still belongs to the same hierarchy of the admissible higher order lattices. In Table 4 we report the minimum T_{min} and maximum temperatures T_{max} , reference temperature T_0 , reduced temperatures θ_{min} and θ_{max} , and the temperature ratio T_{max}/T_{min} .

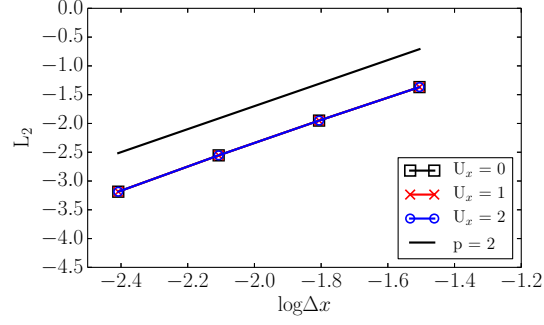


Figure 8: Grid convergence study for the Green-Taylor vortex. Results are shown for non-shifted lattice and lattices with shift of $U = 1$ and $U = 2$.

V_7	T_{min}	T_{max}	T_0	θ_{min}	θ_{max}	T_{max}/T_{min}
$\{0, \pm 1, \pm 2, \pm 3\}$	0.36754	1.63246	0.69795	-0.473397	1.33892	4.44158
$\{0, \pm 1, \pm 2, \pm 4\}$	0.34969	2.71682	1.91103	-0.817014	0.42165	7.76922

Table 4: Temperature range comparison for the original DdQ7^d lattice and its expanded version for increased temperature range.

The temperature ratio T_{max}/T_{min} is the most important parameter to characterize the temperature range. An increase of the temperature range, however, comes at the expense of increased errors in the q_{α}^{eq} and $R_{\alpha\beta}^{eq}$ moments when the Mach number is increased, as shown in Fig. 9.

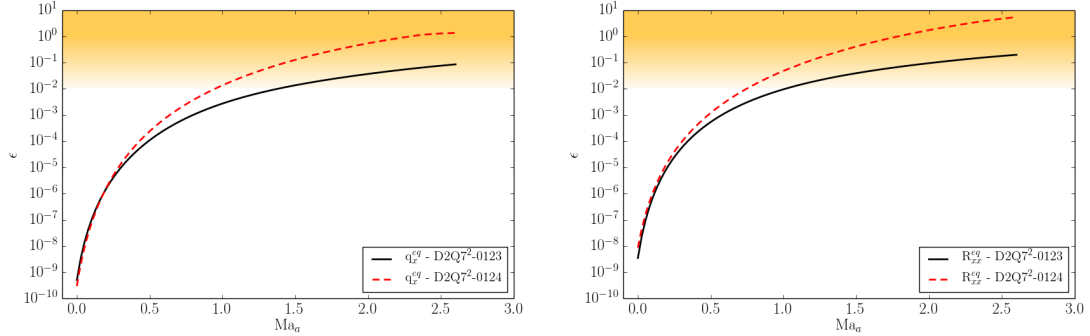


Figure 9: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann counterpart values, as a function of Mach number, for two different lattices: the D2Q7²-0123 and the D2Q7²-0124.

In Figure 10 the same errors are shown as a function of the temperature. At the considered Mach number of $Ma_{loc} = 0.1$ the errors stay small over the entire temperature range for both

the lattices.

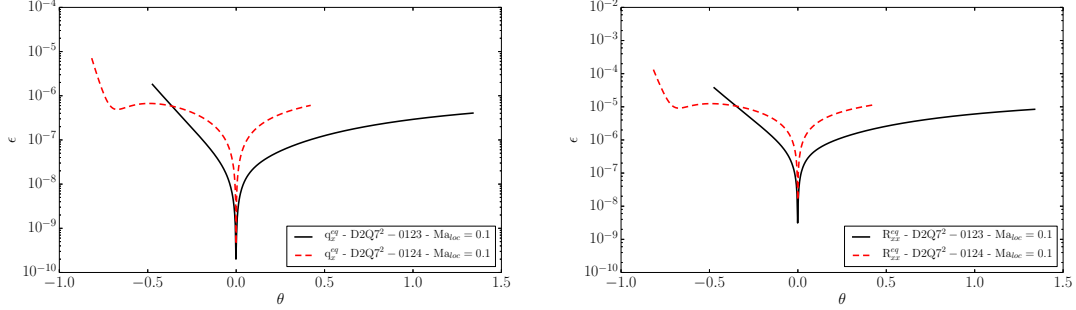


Figure 10: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of the reduced temperature θ , at a local Mach number of $\text{Ma}_{\text{loc}} = 0.1$, and for two different lattices: the D2Q5²-0123 and the D2Q7²-0124.

280

In conclusion, since the errors of the D2Q7²-0124 lattice being higher than for the D2Q7²-0123 lattice, the latter should be the preferred choice. However, it may happen that for certain simulations the temperature range that needs to be achieved is greater than the maximum allowed by the standard $DdQ7^d$ lattice, while the Mach number in the simulation does not reach
 285 supersonic values. In this particular case the D2Q7²-0124 (or D3Q7³-0124) lattice could become particularly useful. We demonstrate the usefulness of these increased temperature range lattice using the simulation of Richtmyer-Meshkov instability in the next section.

4. Numerical results

In this section, we validate the model, the shifted lattices and the lattice with greater temper-
 290 ature range by means of five different set-ups: the supersonic inviscid flow around a diamond airfoil, the flow around the Busemann biplane from subsonic to supersonic Mach numbers, supersonic viscous flow field around a NACA0012 airfoil and finally the interaction of one and two vortices with a shock wave. The increased temperature range lattice is validated by means of the Richtmyer-Meshkov simulation. The boundary conditions presented in [32] are employed for
 295 all the validations.

4.1. Supersonic diamond airfoil

We start by showing, in Fig. 11, the 2D simulation of the supersonic flow field around a diamond shaped airfoil. The inflow Mach number is set to $Ma_{in} = 2.2$, while the Reynolds number is $Re = 10^6$ based on the chord length, and the chord is resolved with $C = 500$ grid points; the wedge angle is $\theta = 12^\circ$. The shifted lattice with $U = 1$ is employed along with slip boundary conditions at the wall. One can notice the characteristic oblique shocks appearing at the leading and

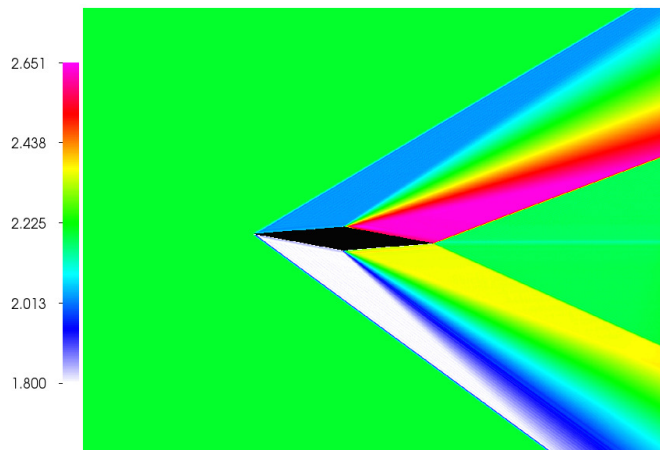


Figure 11: Steady state solution of the Mach number distribution around a diamond shaped airfoil at $Ma = 2.2$, $Re = 10^6$ and $AoA = 3^\circ$.

trailing edges of the airfoil and the expansion wave propagating from the midspan of the airfoil. In Table 5, we compare the results obtained with the ELBM solver for pressure distribution, Mach number distributions and oblique shock angles, with analytical solution employing thin airfoil approximation [42]. Agreement between theory and simulation is excellent, and demonstrates the accuracy of the shifted lattice.

4.2. Busemann biplane

In order to validate the model and the shifted lattices over a wide range of Mach numbers, we show here simulations of the Busemann biplane [43] in its two-dimensional configuration. The Busemann biplane is composed of two half-diamond airfoils (as shown in the insets in Fig. 12), and was originally conceived in order to reduce wave-drag for supersonic flight. It is well known that conventional airfoils experience a dramatic increase of the drag coefficient c_d near sonic conditions, and then, with increasing Mach number, the drag coefficient starts to progressively

		P_1/P_∞	P_2/P_∞	M_1	M_2	β
Up	Analytical	1.1944	0.5616	2.0860	2.5693	29.4033
	ELBM	1.1951	0.5615	2.0859	2.5702	29.6203
	Error	$5.861 \cdot 10^{-4}$	$1.781 \cdot 10^{-4}$	$4.794 \cdot 10^{-4}$	$3.503 \cdot 10^{-4}$	$7.3810 \cdot 10^{-3}$
Down	Analytical	1.6717	0.8334	1.8609	2.3077	34.7915
	ELBM	1.6708	0.8334	1.8581	2.3074	34.7091
	Error	$5.384 \cdot 10^{-4}$	$1.780 \cdot 10^{-4}$	$1.5046 \cdot 10^{-3}$	$1.300 \cdot 10^{-4}$	$2.3684 \cdot 10^{-3}$

Table 5: Comparison between analytical quantities and ELBM results for the diamond airfoil at high Mach problem. Compared are the pressure ratio P_*/P_∞ and the Mach number Ma_* for the first half and the second half of the airfoil, denoted 1 and 2 respectively, and the leading edge oblique shock angle β ; Every quantity for both the upper and the lower surface.

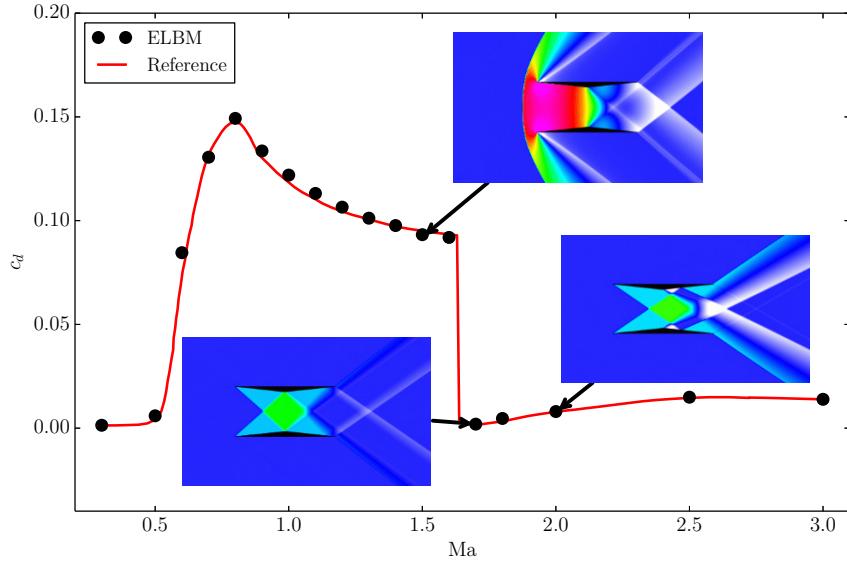


Figure 12: Drag coefficient c_d as a function of the free stream Mach number for the Busemann biplane simulations. Insets represent a snapshot of the pressure distribution around the biplane for three different Mach numbers: $Ma = 1.5$, top; $Ma = 1.7$, bottom left; $Ma = 2.0$, bottom right. The pressure is shown between $p_{min} = 0.5$, white, and $p_{max} = 1.9$, pink, in lattice units.

decrease again. The same happens in the case of the Busemann biplane, where at its simplest
315 configuration, the drag coefficient increases dramatically near sonic conditions, and, similar to
conventional diamond airfoils, it starts to decrease when Mach number is further increased.

However, for the Busemann biplane, after a critical Mach number, the drag coefficient drops well below that of the normal diamond shaped airfoil. This is reflected in Fig. 12, where we compare the drag coefficient computed with the present ELBM model, to the simulations reported in [43].

320 We performed simulations of the Busemann biplane in its simplest configuration, with two symmetric half diamond airfoils with zero angle of attack. The chord of the airfoil was resolved with $C = 400$ grid points. The range of simulated Mach numbers varies between $\text{Ma}_{\min} = 0.3$ to $\text{Ma}_{\max} = 3.0$. Along with the increasing Mach number, different lattices have been employed: from Mach number $\text{Ma}_{\min} = 0.3$ to $\text{Ma} = 1.3$, the conventional symmetric lattice D2Q49 ($U = 0$)
325 was used. From Mach number $\text{Ma} = 1.4$ to $\text{Ma} = 2.0$, the lattice with a shift of $U = 1$ was considered. Finally, for the two highest Mach numbers a lattice with a shift of $U = 2$ in the x -direction was chosen. The results of the ELBM model agree well with the simulations of [43]. In particular, the sensitive drag crises happening between a Mach number of $\text{Ma} = 1.6$ and $\text{Ma} = 1.7$ is well captured. This condition of drag crises, corresponds to free-stream Mach num-
330 ber at which the oblique shock waves developing from the leading edges reattach to the mid-point of the opposite airfoil, thus reflecting nearly onto its own trailing edge.

In Figure 12, together with the plot of the drag coefficient as a function of the free-stream Mach number, the pressure distribution around the biplane is shown at three representative Mach numbers: before, near and after drag crises. At $\text{Ma} = 1.5$, before the drag crises, a detached bow
335 shock is formed in front of the airfoils, which is responsible for a dramatic rise of the pressure which in turn increases wave drag of the biplane. When the Mach is increased sufficiently, the bow shock approaches gradually the leading edge of the airfoils, and finally it gradually transforms in two attached oblique shocks at leading edges of the airfoils. At this point drag dramatically decreases. The minimum drag coefficient is obtained at a Mach number of around $\text{Ma} = 1.7$,
340 which is represented by the lower left inset. As anticipated above, for this case the oblique shocks of the leading edges attain an optimal configuration in terms of drag coefficient. This is due to a combination of the geometric angle of the oblique shock wave and the geometry of the airfoil; at this Mach number, the oblique shock matches the middle edge of the opposite half-diamond airfoil. This has two effects: an influence on the expansion wave formed there, and a reflection of
345 the oblique shock which reaches exactly the trailing edge of the proper airfoil [43]. This perfect matching induces the minimum drag coefficient reported in Fig. 12. When the Mach number is further increased, e.g. like in the case represented by the lower right inset, $\text{Ma} = 2.0$, the oblique shock configuration does not match no more with the geometry of the biplane, and the drag

coefficient starts to increase again. The computation and validation against other simulations of
 350 a sensitive quantity such as the drag coefficient, clearly demonstrates the capability of the model
 to accurately perform simulations over a wide range of Mach numbers. Also the we demonstrate
 here the typical use of the shift velocity: as the mean Mach number of the flow increases, we
 increase the shift velocity thus minimizing the errors arising from the non-optimal use of the
 lattice frame velocity.

355 4.3. Supersonic NACA0012 airfoil

Further validation of the model for high Mach flow simulations is carried out by simulating the
 viscous supersonic flow field around a NACA0012 airfoil with a free stream Mach number of
 $Ma_\infty = 1.5$, Reynolds number $Re = 10^4$ and zero angle of attack $AoA = 0^\circ$. The chord was
 resolved with a uniform grid of $C = 800$ points. Figure 13 shows the temperature distribution
 around the airfoil at a time $t/(C/U_{in}) = 232.3$. One can observe the typical bow shock forming

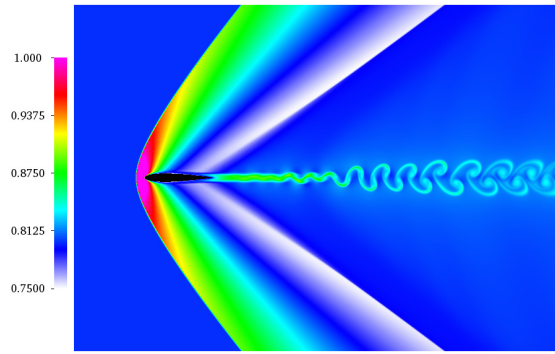


Figure 13: Snapshot of temperature around the NACA0012 airfoil with a free stream Mach of $Ma = 1.5$, a
 Reynolds number $Re = 10^4$ and an angle of attack $AoA = 0^\circ$.

360 in front of the airfoil for supersonic conditions, as well as the the oblique shocks starting as a
 form of a lambda-shock from the trailing edge. Moreover, due to the shear layer developing from
 the trailing edge of the airfoil, vortex shedding is initiated downstream. In order to validate
 the simulation, we plot in Fig. 14 the pressure coefficient along the upstream, airfoil surface and
 365 downstream direction and we compare it with reference [44]. The plot demonstrates the
 capability of the model and shifted lattices to correctly compute the pressure distribution. It
 is clear that introduction of stationary walls is possible also for shifted lattices which assume a
 mean flow. This is due to the presence of a rest particle which resides at the grid node.

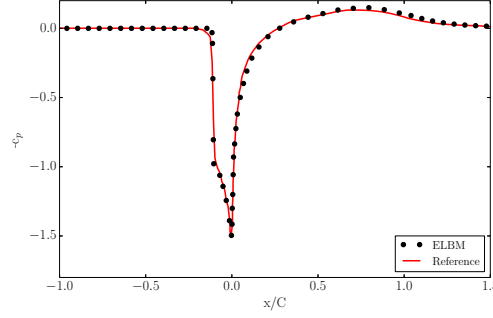


Figure 14: Pressure coefficient in front of the airfoil, on the airfoil surface and behind the airfoil for the simulation of the NACA0012 airfoil at free stream Mach of $Ma = 1.5$, a Reynolds number $Re = 10^4$ and an angle of attack $AoA = 0^\circ$.

4.4. Shock-vortex interaction

In the following, we present simulation of a well studied problem of vortex-shock interaction, where the D2Q7² lattice with a shift of $U=1$ was employed. We compare our solutions to the DNS simulations of [45]. A two-dimensional vortex characterized by a vortex Mach number Ma_v , is advected at a Mach number Ma_a and subsequently interacts with a stationary shock-wave. The advection Mach number corresponds to the relative Mach number of the shock with respect to the upstream velocity, $Ma_s = Ma_a$; the shock is kept at rest. The initial flow field of the vortex is given by:

$$u_\theta(r) = \sqrt{\gamma T} Ma_v r \exp((1 - r^2)/2), \quad (43)$$

and

$$u_r = 0. \quad (44)$$

Above, u_θ and u_r are the tangential and radial velocity of the vortex respectively, and r is the reduced radius $r = r'/R$, with R the characteristic radius. Pressure and density distributions are given by:

$$p(r) = \frac{1}{\gamma} \left[1 - \frac{\gamma-1}{2} Ma_v^2 \exp(1 - r^2) \right]^{\gamma/(\gamma-1)}, \quad (45)$$

$$\rho(r) = \left[1 - \frac{\gamma-1}{2} Ma_v^2 \exp(1 - r^2) \right]^{1/(\gamma-1)}. \quad (46)$$

370 The vortex flow field is superimposed on a constant flow with Ma_a upstream of the shock front. For further details on the numerical set-up the reader is referred to [45]. Three different configurations has been simulated: $Ma_a = 1.2$, $Ma_v = 0.5$ and $Re = 400$; $Ma_a = 1.2$, $Ma_v = 0.25$

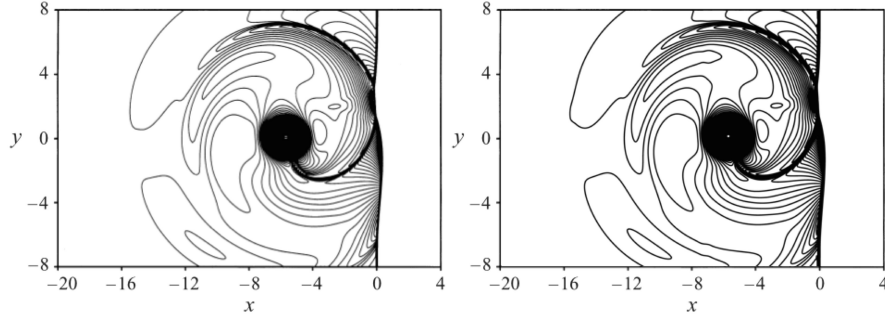


Figure 15: Snapshot of density for both DNS, on the left, and ELBM simulations, on the right, for the case $\text{Ma}_a = 1.2$, $\text{Ma}_v = 0.5$ and $\text{Re} = 400$. Contour levels are from $\rho_{min} = 0.92$ to $\rho_{max} = 1.55$ with an increment of $\Delta\rho = 0.0053$.

and $\text{Re} = 800$; $\text{Ma}_a = 1.05$, $\text{Ma}_v = 0.25$ and $\text{Re} = 400$. The Reynolds number was defined as $\text{Re} = \frac{a_\infty R}{\nu}$, with the speed of sound $a_\infty = \sqrt{\gamma T}$.

In Fig. 15, the isocontours of density ρ for the case $\text{Ma}_a = 1.2$, $\text{Ma}_v = 0.5$ and $\text{Re} = 400$ are compared to the results reported in [45] at a time $t' = t/(R/a_\infty) = 8$. From density contours, one can notice the deformed shock, at $x = 0$, the deformed vortex, at $x \simeq -5$ and $y \simeq 0$, and the reflected shocks developing from the original planar shock, one of which is still connected to the vortex. The density contours show in general a good comparison between DNS and ELBM simulations. In Fig. 16 we compare the radial and tangential sound pressure distribution respectively. The sound pressure is defined as $\Delta p = (p - p_s)/p_s$, where p_s is the pressure behind the shock wave. The radial sound pressure distribution was measured in cylindrical coordinates with origin at the vortex center, at an angle $\theta = -45^\circ$ with respect to the x -axis and at different times $t' = 6$, $t' = 8$, $t' = 10$. The tangential sound pressure distributions were measured in the tangential coordinate at two different radii $r = 6$ and $r = 37$, at time $t = 6$. From the radial distribution one can notice how both the sound precursor (upper sound pressure peak) and the second sound propagate radially from the vortex center with time. Moreover the peak sound Δp_m corresponding to the maximum peak pressure decays with time, for both precursor and second sound. From tangential distribution it is possible to observe the quadrupolar nature of the sound generated by the vortex-shock interaction [45]: one can see in fact, that the second sound, measured at $r = 3.7$ is of opposite sign with respect to the precursor, at $r = 6$. In Fig. 17, we plot the peak sound pressure Δp_m of the precursor, measured at $\theta = -45^\circ$ at different times, and the corresponding radial position. We can remark that the amplitude of the generated sound increases by increasing both shock and vortex Mach numbers Ma_s , Ma_v . To finalize the results

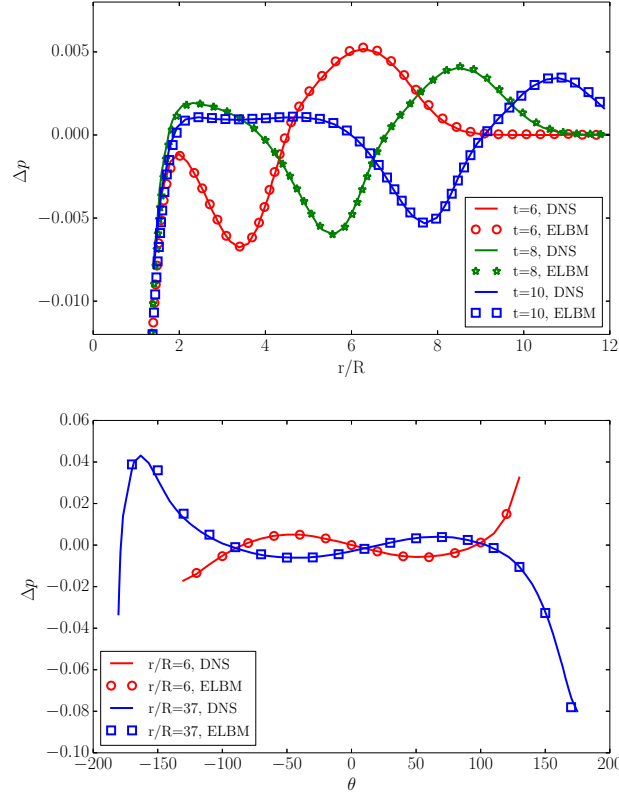


Figure 16: Comparison of the sound pressure distribution Δp in the radial and tangential directions for the case $\text{Ma}_a = 1.2$, $\text{Ma}_v = 0.25$ and $\text{Re} = 800$. Radial sound pressure distribution Δp comparison (top) measured at an angle $\theta = -45^\circ$ with respect to x -axis. Times: $t' = 6$, $t' = 8$, $t' = 10$. Tangential sound pressure distribution Δp comparison (bottom) measured at two different radius $r = 6$ and $r = 3.7$. Time: $t = 6$.

on vortex-shock interaction, we show in Fig. 18 the simulation of a pair of vortices interacting with a shockwave with the same parameters as the simulation presented in Fig. 15: $\text{Ma}_a = 1.2$, $\text{Ma}_v = 0.5$ and $\text{Re} = 400$. We compared the sound pressure of the ELBM simulation and the DNS of [45]. Also in this case the comparison shows good agreement between the DNS and the ELBM method.

Before proceeding with the next set-up, we would like to stress the following point on the non-dimensional time-step of our simulations compared to DNS simulations. The non-dimensional time step for the present ELBM is $\delta t \cdot a_\infty / R = 1.67 \cdot 10^{-2}$ while for the reference DNS it is $\Delta t \cdot a_\infty / R = 1.75 \cdot 10^{-4}$; this shows one of the advantages of the ELBM method over the DNS, requiring two order of magnitude less time steps. Strong coupling of lattice Boltzmann time step

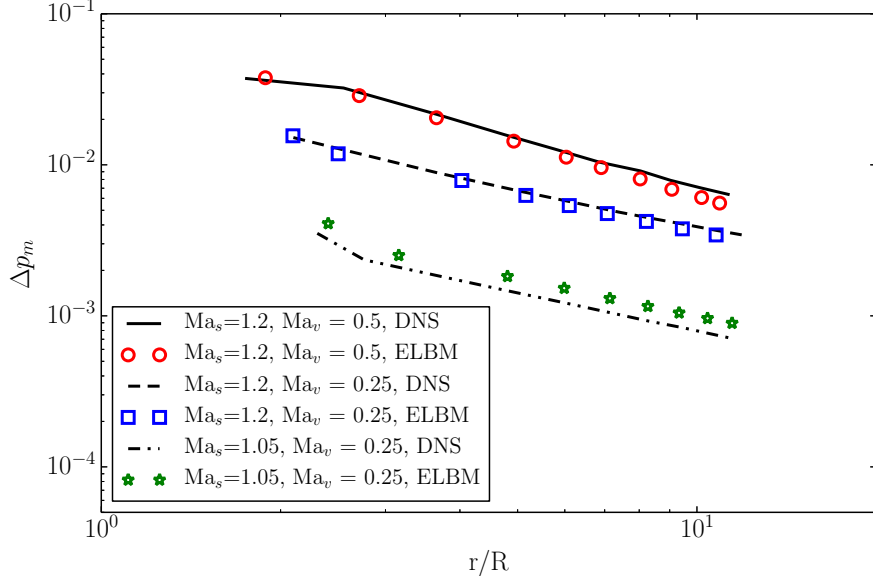


Figure 17: Comparison of decay of peak sound pressure.

405 with the advection velocity or Mach number is a disadvantage for LBM for low Mach number applications; however this becomes an advantage for compressible flows.

4.5. Richtmyer-Meshkov instability

We proceed here with the aim of validating the increased temperature range lattice with the simulation of Richtmyer-Meshkov instability (RMI) [46]. In order to reproduce the experiment
410 of [47] in fact, a minimum temperature ratio of $T_{max}/T_{min} \simeq 4.5$ has to be allowed by the lattice; therefore the standard D2Q7² lattice cannot be used and the increased temperature range lattice needs to be employed.

The RMI problem consists of a shockwave, characterized by a shock Mach number Ma_s , passing through a stratified fluid with a density interface which is of sinusoidal shape with an initial
415 amplitude a_o , and characterized by the pre-shock Atwood number $A = (\rho_2 - \rho_1)/(\rho_1 + \rho_2)$. Where ρ_1 and ρ_2 are the density upstream and downstream of the interface. After the shock front passes the interface, its amplitude grows and the RMI develops. The experimental conditions of [47] have been adopted in the simulation: the incident shock strength was $Ma_s = 1.21$, while the pre-shock Atwood number was $A = (\rho_2 - \rho_1)/(\rho_1 + \rho_2) = 0.601$. The fluid was characterized by
420 a mean adiabatic exponent of $\gamma = 1.276$. The initial interface amplitude is $ka_0 = 0.21$ where k is

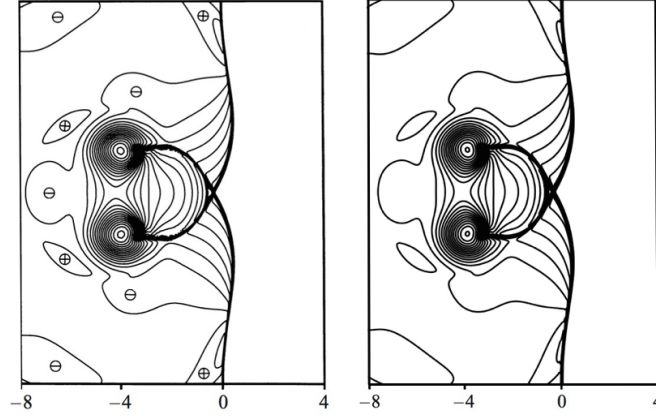


Figure 18: Snapshot of sound pressure for both DNS, on the left ($\oplus$ and $\ominus$ indicate positive and negative sound pressure respectively), and ELBM simulations, on the right, for the case $\text{Ma}_a = 1.2$, $\text{Ma}_v = 0.5$ and $\text{Re} = 400$. Contour levels are from $\Delta p_{\min} = -0.459$ to $\Delta p_{\max} = 0.301$ with an increment of $\Delta p = 0.027$.

the wavenumber of the perturbation $k = 2\pi/\lambda$. For further details the reader is referred to [47]. In the simulation we adopted $\lambda = 1878$ grid points in the transversal direction with respect to the shock motion.

In Fig. 19, we report eight snapshots of the evolution of the density. Time evolves from left to right and from top to bottom. The normalized time τ is defined as $\tau = (t - t_0)ku_0$, where u_0 is the speed of the flow behind the density interface after the shock wave passes through it. The initial conditions are represented by $\tau = -0.03$, where the shock wave is on the left of the sinusoidal density interface. After the passage of the shock, as shown for $\tau = 0.09$, part of the shock travels in the same direction, while the other part is refracted back upstream (see left interface). During the passage of the shock, vorticity is developed instead of deposited is deposited near the density interface, thus initiating the growth of the amplitude of the sinusoidal interface, $\tau = 1.13 - 2.29$. After the initial growth, the instability presents the typical formation of mushrooms-like form, $\tau = 3.44 - 6.92$.

The simulation has been validated by measuring the non-dimensional growth rate of the instability $a(\tau)$ and comparing it with the experimental measurements of [47] and the WENO simulations of [48]. The results are reported in Fig. 20 show the good matching of the ELBM simulation with the experimental results. It should be noted that the WENO results have been non-dimensionalized with the experimental parameters ($u_0 = 628 \text{ cm/s}$).

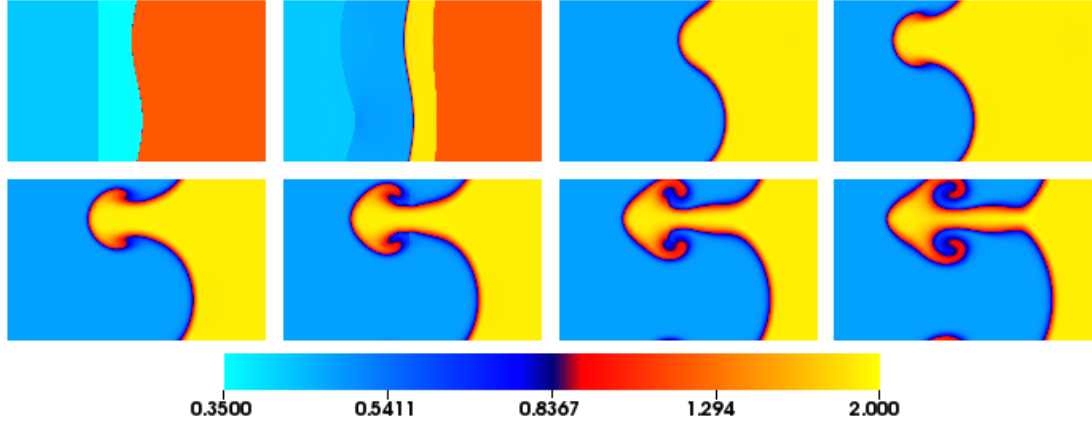


Figure 19: Snapshots of the evolution of the density for the RMI. Time instants from left to right and from top to bottom; first row: $\tau = -0.03$, $\tau = 0.09$, $\tau = 1.13$, $\tau = 2.29$. Second row: $\tau = 3.44$, $\tau = 4.60$, $\tau = 5.76$, $\tau = 6.92$.

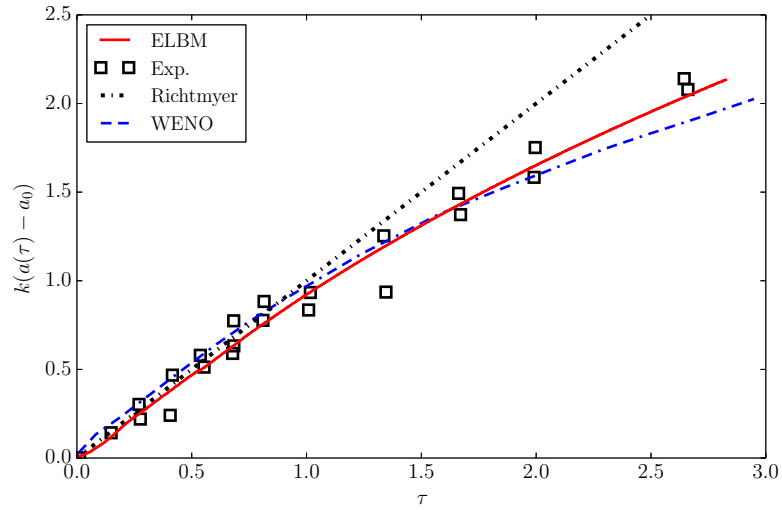


Figure 20: Normalized amplitude growth rate for the Richtmyer-Meshkov instability simulation as a function of normalized time. Plotted are the amplitude evolution for the ELBM, the experiment [47], the WENO simulation [48] and the original analytical prediction of Richtmyer [49] for the initial growth.

5. Conclusions

440 An overview of the main issues arising in the framework of the compressible entropic lattice Boltzmann method are discussed here in detail. In particular, the limited Mach number and the limited temperature range related to a certain lattice are identified as the main weaknesses

of the model. Concerning these issues, we proposed modifications to the standard multispeed lattices, namely by shifting the lattice speeds by one or more unities in a preferential direction, in order to increase the allowed Mach number, and increasing the lattice link length in order to increase the temperature range. It was found, that the weights corresponding to the new class of shifted lattices are in fact, the same weights of the corresponding non-shifted lattice. Also thanks to the recently introduced exact entropic equilibrium, which is computed independently from the lattice velocity set, an increase in the Mach number by a unity or more can be easily achieved just by skewing the lattice velocity set in a preferential direction. The model, together with the novel lattices, were validated by numerous two-dimensional simulations: The supersonic flow field around a diamond-shaped airfoil, subsonic, transonic and supersonic flow field around the Busemann biplane, viscous supersonic flow field around a NACA0012 airfoil, interaction of vortices with a shock wave and the Richtmyer-Meshkov instability. All the benchmark were successfully compared, confirming the capabilities of the compressible ELBM and its extension to higher Mach numbers.

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